Initial Data Collection & Profiling

DAEN 459 – Fall 2026

Alexander Abuabara



Actual x Virtual | Practical x Ritual

The Actual

What currently exists in physical reality. It is concrete, measurable, and historically realized.

The Virtual

The realm of latent potential, mathematical abstraction, and simulations. Real forces or models that shape what can become actual.

The Practical

The bridge of action, utility, and efficiency. It asks: *How do we apply ideas and data to solve problems and make decisions?*

The Ritual

A structured sequence of actions focused on meaning, internal transformation, and alignment rather than external utility.

Everyday Examples

Actual: Our conversation, it's live, it's action.

Virtual: This slides, a book, something that is stored.

Everyday Examples

Actual: Our conversation, it's live, it's action.

Virtual: This slides, a book, something that is stored.

Eating a Meal

- *Practical:* Eating a protein bar at your desk solely to fuel your physical body.
- *Ritual:* Group family dinner, sharing gratitude, and passing specific dishes.

Getting a Degree

- *Practical:* Logging into an administrative portal to download a PDF diploma.
- *Ritual:* Walking across a stage in a cap and gown while your family cheers.

Attribute	The Practical (Utility)	The Ritual (Meaning)
Primary Goal	Efficiency & Output	Transformation & Connection
Focus	The destination or product	The journey and execution
Logic	Instrumental (take the shortcut)	Symbolic (steps cannot be skipped)
Outcome	Changes physical environment	Changes you (psychological/social state)

Data Collection & Profiling

What success looks like

The goal is not "download a dataset"

Produce a **repeatable acquisition path**, a **measured quality baseline**, and a **decision about fitness for purpose**.

1. **Discover:** identify sources, owners, licenses, refresh cadence.
2. **Collect:** land a small but representative raw sample.
3. **Profile:** measure schema, missingness, keys, values, and time.
4. **Decide:** document risks and a go / investigate / reject call.

Artifact: one-page source inventory + raw sample + reproducible profiling report.

Core question: Can we trust this data enough to build on it?

Example: Where do city bikes disappear?

Project idea: forecast bike availability at each station for the next hour.

Source	Fields we want	Risk to test
Trips (CSV)	start/end station, times, trip ID	late uploads; duplicate trips
Station (JSON)	station ID, bikes, docks, timestamp	changing IDs; intermittent readings
Weather (API)	hourly rainfall, temperature	timezone and spatial mismatch

Think ...

A dataset with 10 million trip rows may still be **unfit** for hourly forecasting if it arrives three weeks late.

Collect like an engineer, not a tourist

Source / API $\longrightarrow$ Immutable raw $\longrightarrow$ Profile $\longrightarrow$ Curated tables

Capture provenance

- source URL and access terms
- retrieval UTC timestamp
- requested period and filters
- file hash / batch ID
- API pagination and status

Preserve reproducibility

- keep raw bytes unchanged
- version the extraction code
- record schema and row count
- save failed requests safely
- separate secrets from notebooks

Sample deliberately

Include weekdays *and* weekends, peak *and* quiet hours, old *and* recent batches. A convenient first 1,000 rows can mislead.

Profile across six lenses

Lens	Measure	Example red flag
Structure	fields, types, schema versions	switches type
Completeness	null %, gaps by source/date	30% missing ends
Uniqueness	duplicate IDs, key collisions	repeated `trip_id`
Validity	ranges, allowed values, formats	negative trip duration
Consistency	cross-field / cross-source rules	empty docks $>$ capacity
Freshness	event vs arrival lag, missing hs	API data 8 hours stale

Do not stop at global percentages

Break every important metric down by **date, source, station, and batch**.
An error rate may hide a important broken information.

A miniature profile tells a bigger story

Illustrative sample: 100,000 trip rows across 14 days.

Check	Finding	Interpretation
Null `end_st_id`	8.4%	could be valid free-floating returns
Duplicate `trip_id`	0.7%	test ingestion replay
Duration < 0	0.2%	bad clock or timestamp parsing
Arrival lag > 24 h	12.1%	unsuitable for live features?

Turn metrics into hypotheses

Observation: duplicates spike on Mondays.
Hypothesis: the weekend export is resent each Monday.
Next test: group duplicate IDs by *ingestion batch* before deduplicating.

Time is a data quality dimension

Three different clocks

- **Event time:** when the trip occurred
- **Arrival time:** when our pipeline received it
- **Processing time:** when we transformed it

Check explicitly

- timezone and daylight-saving rules
- late-arriving records
- missing / repeated hourly buckets
- timestamps in the future

Example

A trip at **08:15 local time** may be ingested at **11:40 UTC**. This is not necessarily a 3h25m delay: normalize timezones first.

Capstone rule: define whether each feature uses data *available at prediction time*, not merely data present today.

A small, repeatable profiling script

```
import pandas as pd

# Start with a representative raw sample.
df = pd.read_csv("raw/trips_sample.csv")
report = pd.DataFrame({
    "dtype": df.dtypes.astype(str),
    "null_pct": df.isna().mean().mul(100).round(2),
    "distinct": df.nunique(dropna=True),
})

print("rows:", len(df))
print("duplicate trip IDs:",
      df.duplicated(subset=["trip_id"]).sum())
print(report.sort_values("null_pct", ascending=False))

# Add domain checks, batch-level splits, and time lag tests.
```

Next engineering step: persist the report as a dated artifact and fail CI only on *agreed* thresholds, not arbitrary zero-null expectations.

From finding to action: a quality decision log

Finding	Decision	Evidence needed
0.7% duplicate IDs	quarantine collisions; do not silently delete	compare source rows and batches
8.4% missing end station	treat as "unknown" pending semantics	provider documentation + sample review
12.1% arrive late	exclude from low-latency features	latency distribution over 30 days

Avoid two traps

Trap 1: "clean" means dropping anomalous records.
Trap 2: a dashboard of metrics is a substitute for a decision.

Record **owner, severity, proposed handling, and next verification** for each important issue.

Capstone checkpoint: show your evidence!

Submit four small artifacts:

1. **Source inventory:** URLs, ownership, access, format, cadence, licensing.
2. **Raw landing proof:** file layout, timestamps, hashes, extraction command.
3. **Profiling notebook/report:** schema, 6 quality lenses, segmented metrics.
4. **Decision log:** top 3 risks, tests, mitigations, go/no-go judgment.

90-second project review

"The most surprising data quality issue was _____. It matters because _____. Our next test is _____."

Leave with a verified dataset assumption, not just a downloaded file.